

Numerical Relay

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Introduction

- In utility and industrial electric power transmission and distribution systems, a **digital protective relay** is a computer-based system with software-based protection algorithms for the detection of electrical faults.
- They are functional replacements for electro-mechanical protective relays and may include many protection functions in one unit, as well as providing metering, communication, and self-test functions.

Advantages of Numerical Relay

- Numerical relay consists of many functions in one relay thus replace many traditional relays with one.
- One relay can be used in many ways. Users can configure relay according to their system requirement.
- Consists of no mechanical moving parts hence accurate operation.
- No maintenance is needed like electromechanical relays.
- Relays are compact in size and appearance is good.
- Many complicated functions can be achieved with developed logics using various Gates.
- Gives entire details of fault record including with the graphs.
- Some relays have different Setting groups. The advantage of setting groups is that user can change the entire settings of the relay by simply changing Group setting. Frequent configuration is not needed.
- The main advantage of Numerical relays is communication between relays and other equipment in the network is possible for doing automation of the system.

Numerical Relay



Input processing

- Low voltage and low current signals (i.e., at the secondary of a voltage transformers and current transformers) are brought into a low pass filter that removes frequency content above about 1/3 of the sampling frequency
- The AC signal is then sampled by the relay's analog-to-digital converter
- As a minimum, magnitude of the incoming quantity, commonly using Fourier transform would be used in a simple relay function.
- More advanced analysis can be used to determine phase angles, power, reactive power, impedance, waveform distortion, and other complex quantities.

Logic processing

- The relay analyzes the resultant A/D converter outputs to determine if action is required under its protection algorithm(s).
- Protection algorithms are a set of logic equations in part designed by the protection engineer, and in part designed by the relay manufacturer. The relay is capable of applying advanced logic.
- If a fault condition is detected, output contacts operate to trip the associated circuit breaker(s).

Parameter setting

- The logic is user-configurable and can vary from simply changing front panel switches or moving of circuit board jumpers to accessing the relay's internal parameter setting webpage via communications link on another computer hundreds of kilometers away.
- The relay may have an extensive collection of settings, beyond what can be entered via front panel knobs and dials, and these settings are transferred to the relay via an interface with a PC (personal computer), and this same PC interface may be used to collect event reports from the relay.

Event recording

- In some relays, a short history of the entire sampled data is kept for oscillographic records. The event recording would include some means for the user to see the timing of key logic decisions, relay I/O (input/output) changes, and see, in an oscillographic fashion, at least the fundamental component of the incoming analogue parameters.

Data Display

- Digital/numerical relays provide a front panel display, or display on a terminal through a communication interface. This is used to display relay settings and real-time current/voltage values, etc.
- More complex digital relays will have metering and communication protocol ports, allowing the relay to become an element in a SCADA system. Communication ports may include RS232/RS485 or Ethernet (copper or fibre-optic). Communication languages may include Modbus, DNP3 or IEC61850 protocols.

Numerical Relay front side

Front side

- LED Configuration Menu
- LCD Display
- Numerical Key pad
- Led Reset Button
- Menu Button
- Control Button
- Breaker ON/OFF buttons
- Esc button
- Run LED
- Error LED

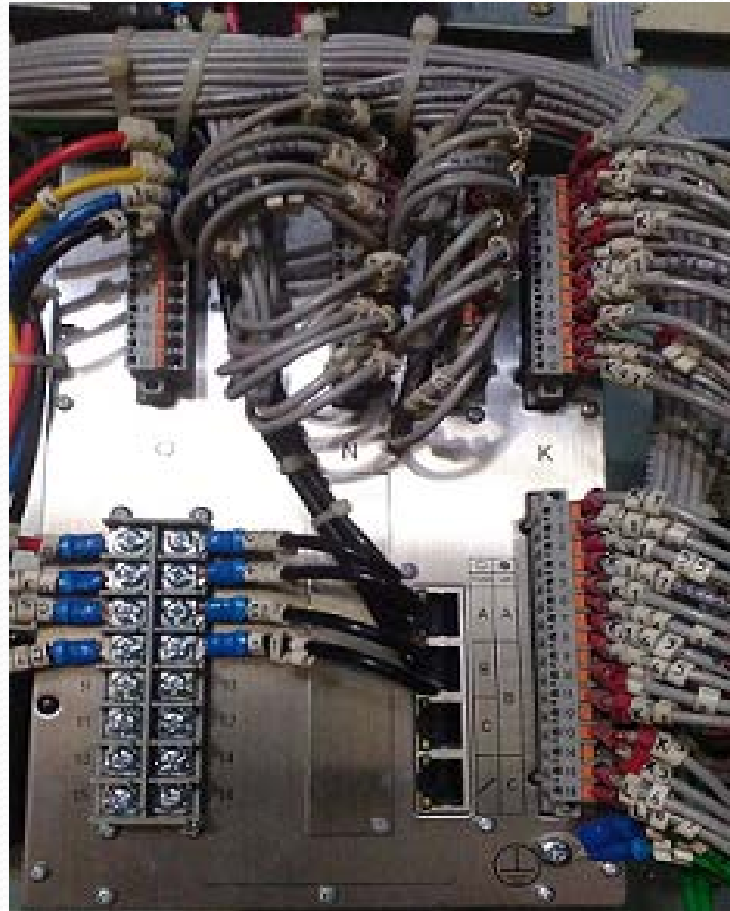


Numerical Relay Back Side:

Relay back side usually consist of different Terminal Blocks [TB] for giving the Digital and Analog inputs, outputs .

- TB for Current Transformer inputs
- TB for Potential Transformer inputs
- TB for Binary Inputs
- TB for Binary Outputs
- TB for Relay Power Supply
- TB for Live status contact
- Slots for Communication Ports

Numerical Relay back side



Input/Output

- **Binary/Digital Inputs:**
- Binary Inputs gives the status of the various equipment to the relay in the form of 1's and 0's, for e.g. the Circuit breaker status is given by BI is either ON/OFF
- **Binary/Digital Outputs:**
- Binary Outputs are the Commands generated by the relay in the form of 1's and 0's, for e.g. trip command given by the relay to breaker in the case of fault.

LED Configuration:

- Any LED can be configured in two ways either Latch or Unlatch.
- **Latch LEDs:**
- The Latch configured LED picks up as soon as the indication exists and remains picked up until the relay is reset at the device.
- The reset has been done by either pressing Reset button manually or via the system interface (SCADA or DCS).
- **Unlatch LEDs:**
- The Unlatch configured LED picks up as soon as the indication exists and drops as soon as the indication no longer applied. It means the LED reset itself and no manual reset required

Communication Protocols:

- Relay communication is provided with the following protocols. These protocols help to communicate relays among them and also with HMI through modem.
- Following are the various protocols used.
- **Modbus RS485:**
- This uncomplicated, serial protocol is mainly used in industry and by power supply corporations.
- **DNP3.0, RS485:**
- DNP 3.0 (Distributed Network Protocol, version 3) is a messaging-based communication protocol.
- **IEC61850 with RJ45 interface or Fibre interface:**
- The Ethernet-based IEC 61850 protocol is the worldwide standard for protection and control systems used by power supply corporations.

Types of Numerical Relays

- **Based on Logic**
- These classifications are made on the basis of logical operation of the relay
- **Over Current/ Earth Fault:** When excessive current flows through a system it will trip the circuit breaker. Used for transformer and feeder protection.
- **Directional overcurrent:** It is operated when the fault drives the power to flow in a particular direction (Opposite to the specified direction). Used in the protection of Bus bar, Generator, and Transformers.

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- **Differential:** The differential relay is set to trip when the phase difference of two or more identical electrical quantities exceeds the specified value. It can Protect Transformers and Generators from localized faults.
 - **Under/ Over Voltage:** The voltage in an electric network might drop or rise below or above a fixed value, the circuit is tripped under such conditions.
 - **Distance:** This type of relay is operated based on the distance between the impedance of the fault and the position of the relay. They are mostly used in the protection of transmission lines.

Based On Characteristics

- Instantaneous relay: Activate the tripping immediately after the occurrence of a fault, there will be no time delay.
- Definite time relay: Activated only if the fault remains until a specific time.
- Inverse time relays with definite minimum time (IDMT): These Relays are mostly used in transmission lines. If the line current exceeds the safe value, circuit breaker gets triggered.
- Voltage restraint over current relay: The relay is activated only if both the under-voltage and overcurrent conditions occur at the same time.

Based on actuating parameters

- Current relays
- Voltage relays
- Frequency relays
- Power relays Etc.

Based on Application

- Primary relay
- Backup relay
- If the protection system fails the whole network might get collapsed so they use the backup relay. Doing this will help us protect the system even if the primary relay goes faulty.

ANSI Device Numbers

- 11 – Multi-function Device
- 21 – Distance
- 24 – Volts/Hz
- 25 – Synchronizing
- 27 – Under Voltage
- 32 – Directional Power Element
- 46 – Negative Sequence Current
- 40 – Loss of Excitation
- 47 – Negative Sequence Voltage
- 50 – Instantaneous Overcurrent (N for neutral, G for ground current)

ANSI Device Numbers

- 51 – Inverse Time Overcurrent (N for neutral, G from ground current)
- 59 – Over Voltage
- 62 – Timer
- 64 – Ground Fault (64F = Field Ground, 64G = Generator Ground)
- 67 – Directional Over Current (typically controls a 50/51 element)
- 79 – Reclosing Relay
- 81 – Under/Over Frequency
- 86 – Lockout Relay / Trip Circuit Supervision
- 87 – Current Differential (87L=transmission line diff; 87T=transformer diff; 87G=generator diff)